

Note: Please answer all the questions in English. Your grade is determined by the minimum of the points you obtained and 30.

Problem 1. (4 points)

Given $\mathbf{P}(A) = 1/3$, $\mathbf{P}(B) = 1/4$, $\mathbf{P}(A \cap B) = 1/6$, find the following probabilities:

a) $\mathbf{P}(A^c \cup B)$

b) $\mathbf{P}(A^c \cap B^c)$

Justify your steps utilizing axioms of probability.

Problem 2. (6 points)

In a row of $N + M$ seats N male students and M female students sit at random. Find the expected number of successive pairs of seats in which a male student and a female student sit.

Problem 3. (7 points)

Let X and Y be discrete random variables with joint mass function

$$f(x, y) = \frac{C}{(x + y - 1)(x + y)(x + y + 1)}, \quad x, y = 1, 2, 3, \dots$$

a) Calculate C .

b) Find the marginal mass functions of X and Y .

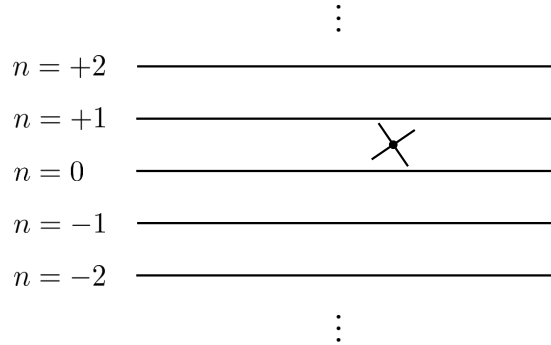
Hint:

$$\frac{1}{(x + y - 1)(x + y)(x + y + 1)} = \frac{1}{2(x + y - 1)(x + y)} - \frac{1}{2(x + y)(x + y + 1)}$$

Problem 4. (7 points)

The plane is ruled by the lines $y = n$, for $n = 0, \pm 1, \dots$, and on to this plane we drop a cross formed by welding together two **unit** needles perpendicularly at their midpoints. Let I and J be the numbers of intersections of the first and second needle with the grid of parallel lines, respectively. Then $Z = I + J$ represents the number of intersections for the entire cross.

- a) Find $\mathbf{E}[Z]$.
- b) Find $\text{var}(Z)$.



Problem 5. (8 points)

Assume that X is a continuous uniform random variable on $[0, 4]$. Y is an exponential random variable, independent from X , with expectation $\mathbf{E}[Y] = 1/2$.

- a) Find the probability that $Y \geq X$.
- b) Find $\mathbf{E}[Z|Y = y]$ where $Z = X + Y$.
- c) A random variable W is defined as follows. We toss a fair coin (independent from X and Y). If the result is "head", we let $W = Y$; if it is tail, we let $W = Y + 2$. Find the probability of "head" given that $W = 3$.